Scalability: From 1 to N Projects

Scalability: From 1 to N Projects I

The Standalone Project Paradigm enables horizontal scaling: adding a new project requires creating a directory with `manuscript/config.yaml` and nothing else. No infrastructure code changes, no `pyproject.toml` modifications, no CI configuration updates. The `run.sh` orchestrator automatically discovers new projects and presents them in its interactive menu.

We have validated scaling with 25 canonical exemplars under `projects/templates/`—always present for onboarding and tooling—and with this manuscript from `projects/templates/template_template` (152 tests) as a git-tracked public exemplar in the same automated discovery menus.

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Canonical trio:

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- **`template_code_project`**: Numerical optimization example with gradient-descent narration and 263 discovered tests. Its `pyproject.toml` declares the 90% project-source floor. Minimal footprint: compact `src/`, scripted analysis, short manuscript sections.
- **`template_prose_project`**: Prose-heavy manuscript emphasizing narrative structure and bibliography discipline, with 165 discovered tests and the same declared project-source floor; tests exercise rendering and Markdown integrity without heavyweight numerics.
- **`template_autoresearch_project`**: AutoResearch readiness workflow — bounded ML task loop (deterministic MNIST run), plan validation (`infrastructure.autoresearch`), evidence-registry reports, and manuscript hydration; its own scripts (`run_autoresearch_loop.py`, `z_generate_manuscript_variables.py`) stay thin over `src/` logic. **`template_search_project`** exercises literature-search workflows (remote corpus JSON fusion, `y_generate_search_figures.py`, review reports).

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Meta manuscript (**`projects/templates/template_template`**) analyzes the repository via `src/template_template/` introspection metrics; it now lives alongside the other public exemplars under `projects/templates/`.

These workspaces share no project-level code—only Layer 1 (31 infrastructure subdirectories, ~1307 Python files)—validating insulation between domain repos and reusable services.

Multi-Project Orchestration I

When the `--all-projects` flag is passed to `run.sh`, the pipeline executes each discovered project sequentially, running infrastructure tests once at the start and skipping them for individual projects to avoid redundant validation. After all projects complete, a cross-project executive report aggregates metrics (test counts, coverage percentages, page counts, rendering durations) into a unified dashboard with both JSON and Markdown output formats. This executive reporting stage provides repository-level visibility without requiring any project-specific reporting code.

Scaling Metrics I

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Metric	template_code_project	template_prose_project	template_autoresearch_project
Source modules	25	8	59
Test files	16	13	23
Test count	263	165	318
Manuscript chapters	9	8	6
Analysis scripts	9	5	5
Figures (auto-generated)	9	5	27

Scaling Metrics I

The infrastructure overhead per project is constant regardless of project size: the same 31 modules, the same 14 pipeline stages, the same rendering and validation logic. This O(1) infrastructure cost is the architectural payoff of the Two-Layer separation.